Math Kangaroo Practice

Grade 3-4 · 20 questions · standard · seed 41187

Part A: Each question is worth 3 points

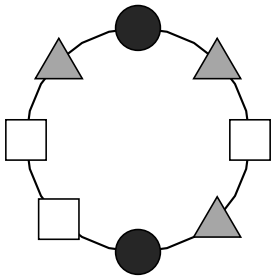
1. 48 children stand in a row. Each child wears a T-shirt and holds a card with a shape on it. The T-shirts repeat like this: black, white, black, white, ... The cards repeat like this: triangle, circle, star, triangle, circle, star, ... How many children have a **black** T-shirt **and** a **triangle** card?

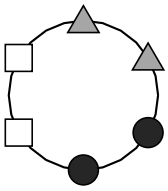
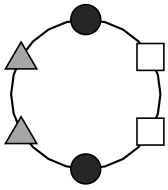
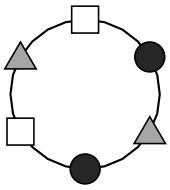
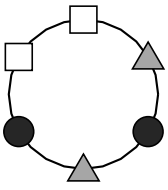
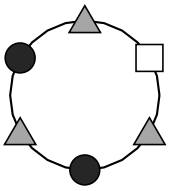
A. 4 ***B.*** 5 ***C.*** 6 ***D.*** 7 ***E.*** 8

2. This tray of cookies lies in the kitchen. Some cookies are dark (chocolate), some are white. Three girls take cookies from the tray, one after the other:
Zoe takes all the **heart-shaped** cookies that are still on the tray.
Nora takes all the **white** cookies that are still on the tray.
Ida takes all the **large** cookies that are still on the tray.
First Ida goes, then Nora, then Zoe. How many cookies does **Zoe** take?

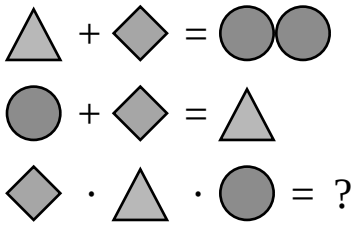
A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 6

3. The picture shows a bracelet with circles, squares and triangles. Anna removes two neighbouring stones: one square and one triangle. Which bracelet can Anna get?



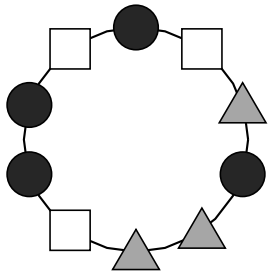
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

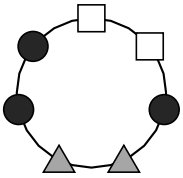
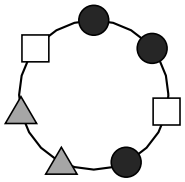
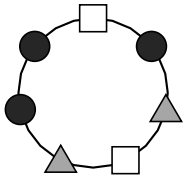
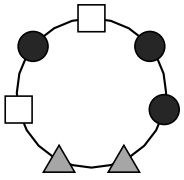
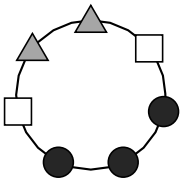
4. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Mia lays out the cards so that the calculations are correct. What number does Mia get for the last calculation?



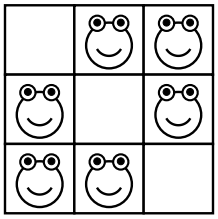
A. 30 ***B.*** 31 ***C.*** 35 ***D.*** 37 ***E.*** 60

5. The picture shows a bracelet with circles, squares and triangles. Lisa removes three neighbouring stones, one of each shape. Which bracelet can Lisa get?



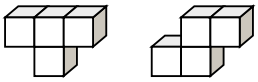
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

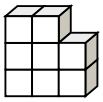
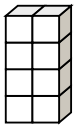
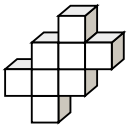
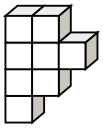
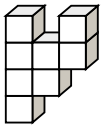
6. There are exactly 2 frogs in each row and in each column of this 3 × 3 pond. A frog can jump to a **neighbouring** field (left, right, up or down) if it is empty. Only **one** frog jumps. How many different jumps are possible? (Count every frog and every empty field it can reach.)



A. 3 ***B.*** 5 ***C.*** 6 ***D.*** 7 ***E.*** 8

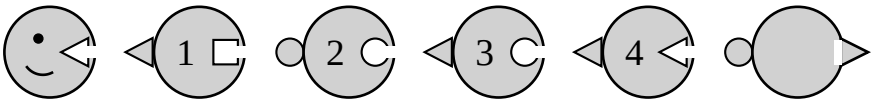
7. Kanga wants to build a flat figure out of these 2 parts made of cubes (the picture). He must use every part once, and he can turn the parts or flip them over. Which of these figures **can** he make?



A.  ***B.***  ***C.***  ***D.***  ***E.*** 

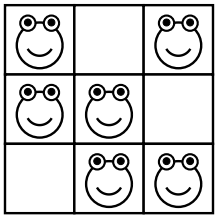
Part B: Each question is worth 4 points

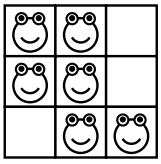
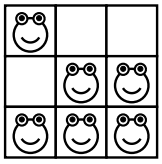
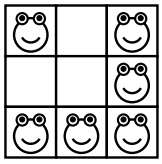
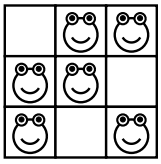
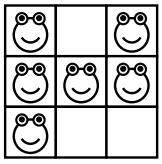
8. Emma has these pieces: a head, a tail and 4 parts with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A caterpillar is the head, then 1, 2, 3 or 4 parts, then the tail. Each piece can be used only once. Which part can **never** be part of one of Emma's caterpillars?



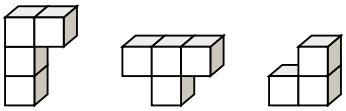
A. part 1 ***B.*** part 2 ***C.*** part 3
D. part 4 ***E.*** each part can be used

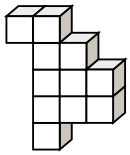
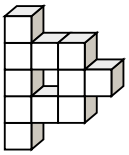
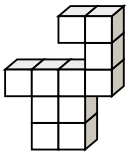
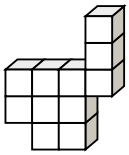
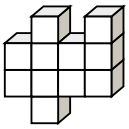
9. There are exactly 2 frogs in each row and in each column of this 3 × 3 pond. A frog can jump to a **neighbouring** field (left, right, up or down) if it is empty. Two frogs jump at the same time, each to an empty neighbouring field. After that there are again 2 frogs in each row and in each column. Which picture can show the pond **after** the jump?



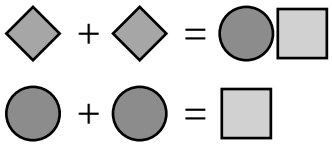
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

10. Kanga wants to build a flat figure out of these 3 parts made of cubes (the picture). He must use every part once, and he can turn the parts or flip them over. Which of these figures can he **not** make?



A.  ***B.***  ***C.***  ***D.***  ***E.*** 

11. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Sara lays out the cards so that the calculations are correct. Which two-digit number do the cards square and diamond make?



A. 8 ***B.*** 21 ***C.*** 23 ***D.*** 25 ***E.*** 26

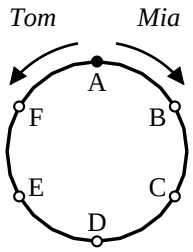
12. This tray of cookies lies in the kitchen. Some cookies are dark (chocolate), some are white. Three girls take cookies from the tray, one after the other:
Rosa takes all the **heart-shaped** cookies that are still on the tray.
Lena takes all the **white** cookies that are still on the tray.
Mia takes all the **large** cookies that are still on the tray.
We do not know in which order they go. How many cookies are still on the tray after all three girls have taken theirs?

A. 4 ***B.*** 5 ***C.*** 6 ***D.*** 7 ***E.*** 8

13. 60 children stand in a row. Each child wears a T-shirt and holds a card with a shape on it. The T-shirts repeat like this: dotted, black, dotted, black, ... The cards repeat like this: star, cross, diamond, star, cross, diamond, ... What does child number **52** wear?

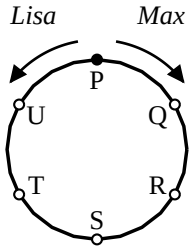
A. black T-shirt, cross card ***B.*** dotted T-shirt, star card ***C.*** black T-shirt, star card
D. black T-shirt, diamond card ***E.*** dotted T-shirt, cross card

14. Mia and Tom start at the same time from point A of the circle. Mia runs clockwise, Tom runs anticlockwise (see picture). They meet for the first time at point E and keep on running. How many times do they meet until they are together at point A again? (Count the meeting at A, but not the start.)



A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 6

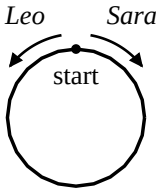
15. Max and Lisa start together at point P and run around the circle in opposite directions: Max clockwise and Lisa anticlockwise (see picture). The points are evenly spaced. Max runs 2 times as fast as Lisa. At which point do they meet for the first time?



A. point P ***B.*** point Q ***C.*** point R ***D.*** point T ***E.*** point U

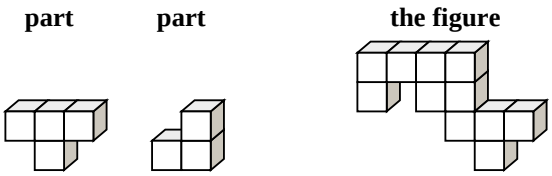
Part C: Each question is worth 5 points

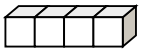
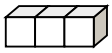
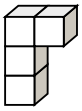
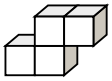
16. Sara and Leo start together at the same spot of a round track. Sara runs clockwise and Leo runs anticlockwise. Sara needs 12 minutes to run once around, Leo only needs 4 minutes. After 24 minutes they stop — both are back at the start together. How many times did they meet? (Count the meeting at the end, but not the start.)



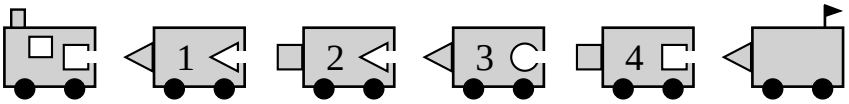
A. 4 ***B.*** 6 ***C.*** 7 ***D.*** 8 ***E.*** 9

17. Kanga built this flat figure out of 3 parts made of cubes, turned or flipped. Two of the parts are shown next to it. Which part is the third one?



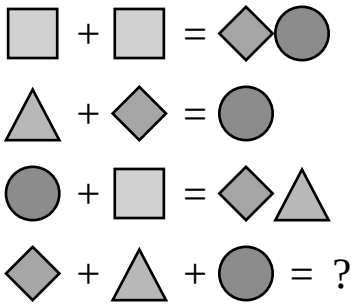
A.  ***B.***  ***C.***  ***D.***  ***E.***

18. Lucas has these pieces: an engine, a last wagon and 4 wagons with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A train is the engine, then 1, 2, 3 or 4 wagons, then the last wagon. Each piece can be used only once. Lucas has already put wagon 4 right after the engine. In how many different ways can Lucas finish the train?



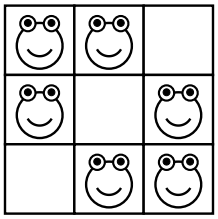
A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

19. Cards of the same shape hide the same digit, and cards with different shapes hide different digits. Two cards side by side make a two-digit number. Tom lays out the cards so that the calculations are correct. What number does Tom get for the last calculation?



A. 16 ***B.*** 17 ***C.*** 18 ***D.*** 20 ***E.*** 25

20. There are exactly 2 frogs in each row and in each column of this 3 × 3 pond. A frog can jump to a **neighbouring** field (left, right, up or down) if it is empty. Two frogs jump at the same time, each to an empty neighbouring field, and after that there are again 2 frogs in each row and in each column. How many of the frogs can **never** be one of the two jumping frogs?



A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 6

Answer Key

1.	**E**	**2.**	**B**	**3.**	**D**	**4.**	**A**	**5.**	**B**	**6.**	**E**
7.	**D**	**8.**	**A**	**9.**	**D**	**10.**	**D**	**11.**	**E**	**12.**	**A**
13.	**C**	**14.**	**C**	**15.**	**D**	**16.**	**D**	**17.**	**C**	**18.**	**B**
19.	**A**	**20.**	**C**								

Solutions

1. E (8) — Both patterns start again together every 6 children. In each block of 6 only child number 1 (then 7, 13, ...) has a black T-shirt and a triangle card. 48 ÷ 6 = 8 blocks, so 8. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · both_count]

2. B (1) — Cross out what the girls before Zoe take first. Of her cookies only 1 are still on the tray (she would get 3 if she went first). [set-01/by_level/Grade_03-04_Ecolier/2024 Q24 · given_order]

3. D — Go around the bracelet and look at every group of 2 neighbours (also the ones across the top). Taking away the group square, triangle leaves the bracelet in D, turned a bit. The other pictures need stones that were not next to each other, or other stones. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · remove_two]

4. A (30) — Two digits add up to at most 9 + 9 = 18, so the circle at the front of the two-digit answer must be 1. Try the possible digits and check every calculation: triangle = 6, circle = 1, diamond = 5. So 5 · 6 · 1 = 30. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · product_all]

5. B — Go around the bracelet and look at every group of 3 neighbours (also the ones across the top). Taking away the group circle, square, triangle leaves the bracelet in B, turned a bit. The other pictures need stones that were not next to each other, or other stones. [set-01/by_level/Grade_03-04_Ecolier/2025 Q21 · remove_three]

6. E (8) — Go frog by frog and count its empty neighbours: 2 + 0 + 2 + 2 + 0 + 2 = 8. [set-01/by_level/Grade_03-04_Ecolier/2024 Q21 · one_frog]

7. D — Look for one part inside the figure; what is left must be the other part. That works only for figure D. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · two_parts]

8. A (part 1) — Start at the head and try every piece that fits, one branch at a time (H = head, T = tail): H–3–T, H–3–2–T, H–4–3–T, H–4–3–2–T. That is 4. Part 1 is in none of them: it is a dead end, or nothing leads to it. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · never_used]

9. D — First count the frogs in every row and column of each picture. Then look at which frogs moved: in the right picture each of the two frogs moved just one field (not diagonally), and the second jump undoes the first. [set-01/by_level/Grade_03-04_Ecolier/2024 Q21 · which_picture]

10. D — Try to fill each figure with the parts, starting at a corner or a cube that sticks out. Four figures can be filled; in figure D there is always a gap that no part fits. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · cannot_make]

11. E (26) — Two digits add up to at most 9 + 9 = 18, so the circle at the front of the two-digit answer must be 1. Try the possible digits and check every calculation: diamond = 6, square = 2, circle = 1. So the square is the tens digit and the diamond the ones digit: 26. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · plus_minus]

12. A (4) — A cookie stays only if it is not a heart, not white and not large: the small dark round ones. There are 4 of them - the order does not matter. [set-01/by_level/Grade_03-04_Ecolier/2024 Q24 · left_over]

13. C (black T-shirt, star card) — The T-shirts repeat every 2: 52 = 25 × 2 + 2, so the T-shirt is the 2nd one: black. The cards repeat every 3: 52 = 17 × 3 + 1, so the card is a star. [set-01/by_level/Grade_03-04_Ecolier/2024 Q19 · child_number]

14. C (3) — Tom moves 2 letters on from one meeting to the next: E, C, A. That is 3 meetings. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · skip_points]

15. D (point T) — Until they meet they run once around the circle together, and Max runs 2 times as far: Lisa runs 1 part and Max 2 parts of 3. 1 part of 3 is 2 of the 6 gaps, so Lisa only gets to point T.
[set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · where_first]

16. D (8) — Between two meetings they run once around the track together. In 24 minutes Sara runs 2 and Leo runs 6 times around: 2 + 6 = 8 laps together, so 8 meetings. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · minutes]

17. C — The figure has 11 cubes and the two parts have 7, so the third part has 4 cubes. Put the two parts into the figure; what is left has the shape of part C. [set-01/by_level/Grade_03-04_Ecolier/2024 Q23 · missing_part]

18. B (2) — Look only at the branch that starts with wagon 4: H–4–2–T, H–4–2–1–T. That is 2.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · start_given]

19. A (16) — Two digits add up to at most 9 + 9 = 18, so the diamond at the front of the two-digit answer must be 1. Start with the calculation where the fewest digits are possible, then check all three: square = 9, circle = 8, diamond = 1, triangle = 7. 1 + 7 + 8 = 16. [set-01/by_level/Grade_03-04_Ecolier/2024 Q20 · four_cards]

20. C (2) — Find all the ways (4): the second jump must undo the first one (sideways between the same two columns, or up/down between the same two rows). 4 frogs take part in some way, so 6 - 4 = 2 never jump.
[set-01/by_level/Grade_03-04_Ecolier/2024 Q21 · never_jumps]